Chapter Notes: The Amazing World of Solutes, Solvents, and Solutions

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Have you ever wondered why sugar completely disappears when you stir it in water, but sand does not?

Welcome to **The Amazing World of Solutes, Solvents, and Solutions**! In this chapter you will learn what happens at the microscopic level when substances mix, why some mixtures become clear while others remain cloudy, and how solutions - from fruit juices to seawater - are part of everyday life. We shall study definitions, types of solutions, solubility, the effect of temperature, the behaviour of gases in liquids, and the ideas of density and buoyancy with practical examples and simple experiments.

What are Solute, Solvent and Solution?

What is a solution?

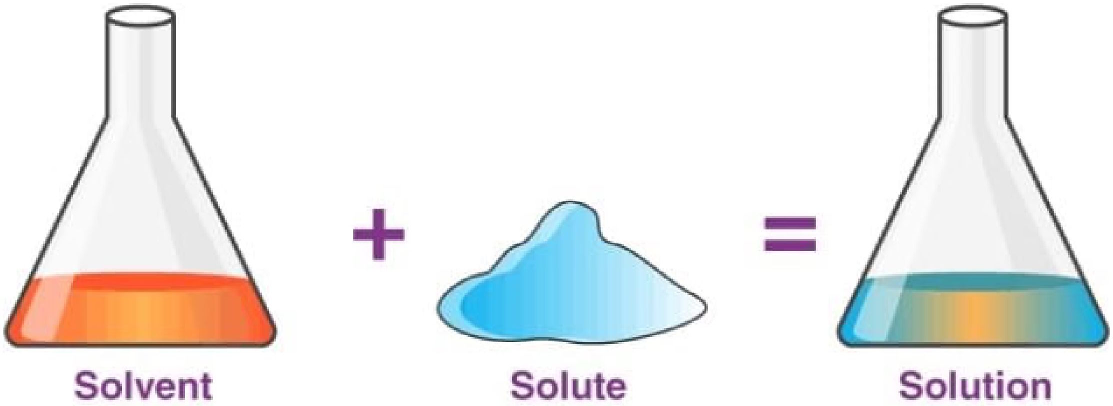
A **solution** is a **uniform (homogeneous) mixture** in which the components are evenly distributed and cannot be seen separately.

Example: When salt or sugar is mixed with water and dissolves completely, the result is a clear solution.

When sand or sawdust is mixed with water, the solid particles do not dissolve and either float or settle - this is not a solution.

Basic relation:

$$\text{Solute} + \text{Solvent} = \text{Solution}$$



Key definitions

Solute: The substance that dissolves in the solvent. In many solid-liquid solutions the solid (for example, salt or sugar) is the solute and is usually present in smaller amount.

Solvent: The substance that dissolves the solute. The solvent is usually present in larger amount and determines the physical state of the solution (for example, water in saltwater).

Solution formation: When solute particles disperse uniformly among solvent particles so that individual components cannot be seen or separated by simple mechanical methods.

Types of Solutions and Examples

Solid in liquid

Solute: Solid (e.g., salt, sugar).

Solvent: Liquid (e.g., water).

Example: Saltwater or sugar water - clear and homogeneous.

Liquid in liquid

When two liquids mix to form a homogeneous mixture, the one present in smaller quantity is called the solute and the larger one the solvent.

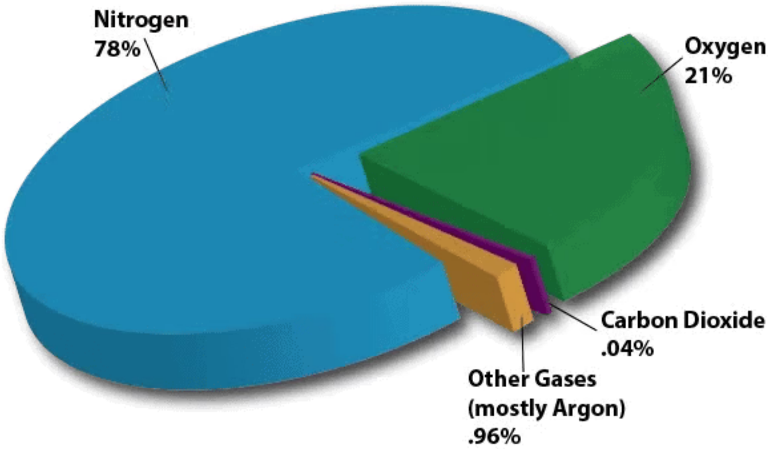
Example: Vinegar is a solution of acetic acid in water; acetic acid is the solute and water the solvent.

Gas in gas (gaseous solutions)

Gases can form solutions. **Air** is a common example.

Solvent: Nitrogen (about 78% of air).

Solutes: Oxygen, argon, carbon dioxide and other gases in smaller amounts.



Air is a mixture of Gases

Special cases and interesting facts

Even if the solute amount is larger than the solvent amount, the liquid is still treated as the solvent because it is the medium in which the solute dissolves. For example, in **chashni** (sugar syrup) a large amount of sugar dissolves in a small amount of water; water remains the solvent.

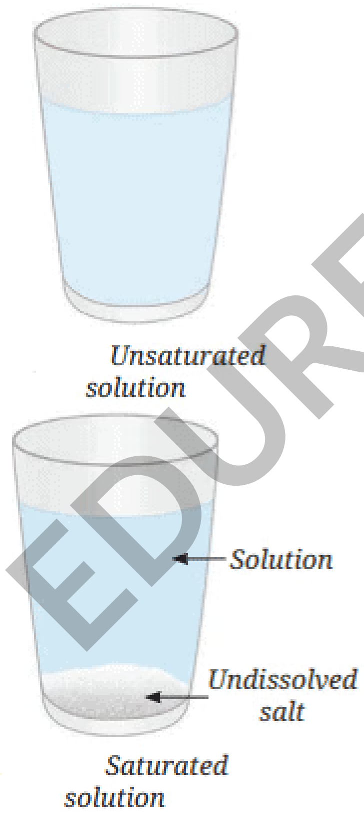
Solubility depends on the nature of both the solute and the solvent, and on conditions such as temperature and pressure.

How much solute can a fixed amount of solvent dissolve?

A simple salt-in-water experiment shows that a solvent has a finite capacity to dissolve a solute. The experiment also introduces the terms **unsaturated**, **saturated**, **concentrated** and **dilute**.

Take a clean glass tumbler half-filled with water.

Add a spoonful of salt and stir until dissolved.
Keep adding spoonfuls and stir each time until added salt no longer dissolves and settles at the bottom.

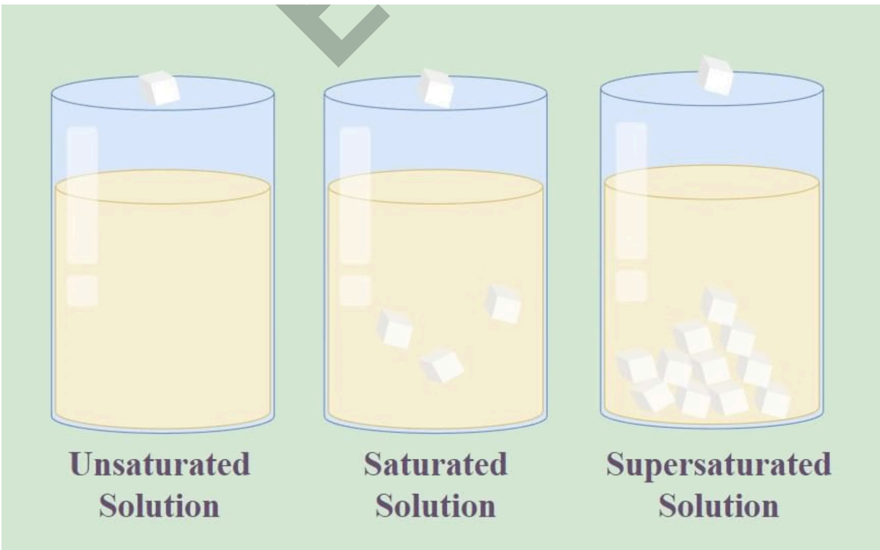


Observations and inferences

At first the salt dissolves completely and the solution remains clear.
After a certain number of spoonfuls (the exact number depends on the amount of water and its temperature) further salt does not dissolve and settles on the bottom.
This shows that the solvent has reached its dissolving limit for that temperature and amount - the solution is then **saturated**.

Important concepts

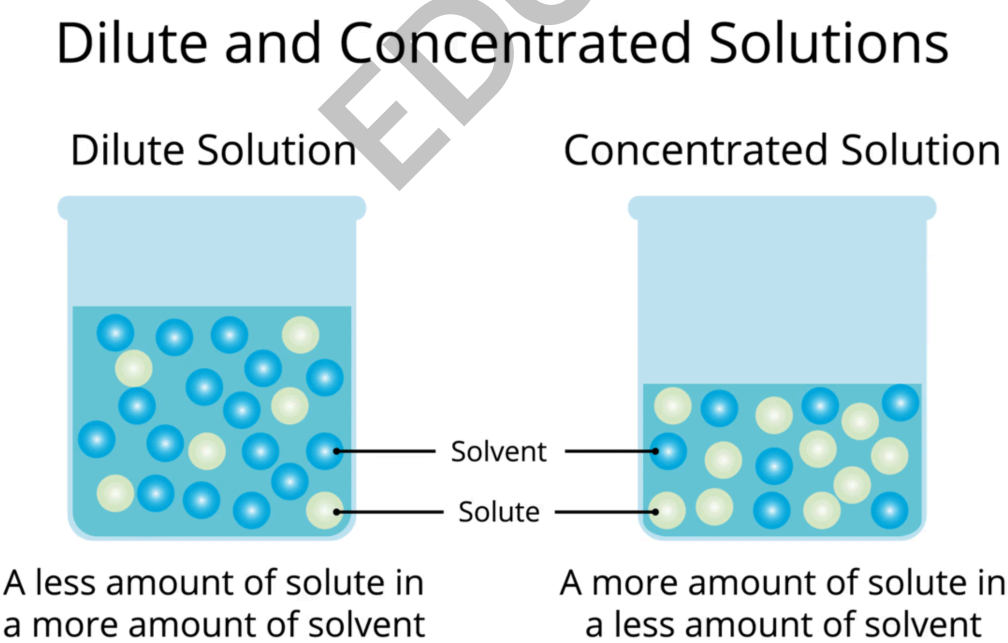
Unsaturated solution: A solution that can still dissolve more solute at the given temperature (for example, when only 1-2 spoonfuls of salt are dissolved).
Saturated solution: A solution that cannot dissolve any more solute at the given temperature; excess solute remains undissolved and settles.



Concentration: The amount of solute present in a given quantity of solution (or solvent). It tells us how 'strong' a solution is.

Dilute solution: Contains a relatively small amount of solute (for example, one spoon of salt in a large volume of water).

Concentrated solution: Contains a relatively large amount of solute (for example, many spoons of salt in a small volume of water).



Solubility: The maximum amount of solute that can dissolve in a fixed amount of solvent at a given temperature (often quoted as grams per 100 mL of solvent at a specified temperature).

MULTIPLE CHOICE QUESTION

Try yourself: What happens when you add more salt to water after reaching its limit?

A It changes color.

B It forms bubbles.

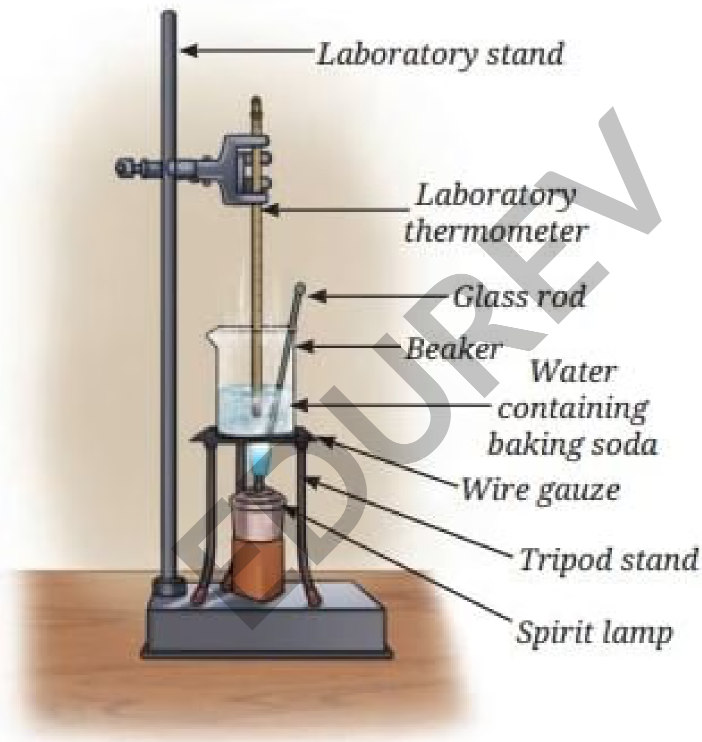
C It settles at the bottom.

D It dissolves completely.

View Solution

Effect of temperature on solubility

Temperature often affects how much solute a solvent can dissolve. For most solid solutes in liquids, solubility increases with temperature; for gases dissolved in liquids, solubility usually decreases as temperature rises.



Dissolution of baking soda in water

Experiment (heating increases solubility of many solids)

Take 50 mL of water in a beaker and note its temperature (for example, 20 °C).
Add small amounts of baking soda, stirring until it dissolves each time, until some remains undissolved.
Heat the mixture to about 50 °C while stirring; observe that previously undissolved baking soda dissolves.
Heat further (for example to 70 °C) and note that even more baking soda dissolves before saturation occurs.

Observations and inferences

At higher temperatures the same amount of solvent can dissolve more of many solid solutes.
Heating a saturated solution can often make it unsaturated, allowing more solute to dissolve.
This practical fact is used in processes such as recrystallisation in the laboratory and in preparing concentrated sugar syrups.

Solvents in traditional Indian medicine

Water is the principal solvent used to prepare many medicinal formulations in Ayurveda, Siddha and other traditional systems.

Hydro-alcoholic extracts are made by using mixtures of water and alcohol to extract active compounds from herbs.
Other solvents used in traditional formulations include oils, ghee and milk; they help extract and deliver fat-soluble compounds and enhance therapeutic action.

Be a scientist: Asima Chatterjee

Inspiration: Asima Chatterjee carried out research on medicinal plants, isolating useful compounds by using solvents and solution techniques.

Asima Chatterjee

Achievements:

- Developed plant-based **anti-epileptic** and **anti-malarial** drugs.
- Awarded a **Doctorate of Science** (one of the earliest Indian women to receive it).
- First woman recipient (in chemical sciences) of the **Shanti Swarup Bhatnagar Award**.
- Honoured with the **Padma Bhushan**.

Solubility of gases

Many gases, including **oxygen**, dissolve in water, though only in small amounts; the dissolved oxygen is vital for aquatic life.
The mixture of a gas dissolved in a liquid is homogeneous - the gas molecules are distributed uniformly through the solvent.

Aquatic species in water

Effect of temperature on gas solubility

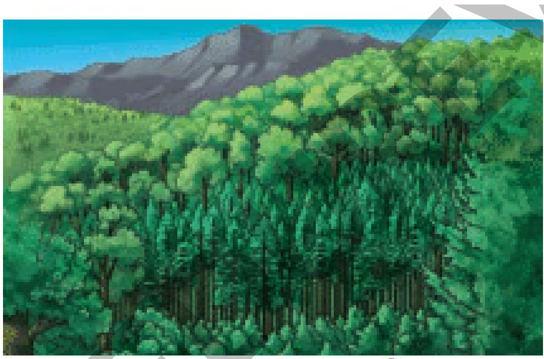
For most gases the solubility in a liquid **decreases** with increasing temperature.

Example: Cold water can hold more dissolved oxygen than warm water; this is important for fish and other aquatic organisms.

Why do objects float or sink in water?

Objects float or sink depending mainly on **density**, a measure of how much mass is present in a given volume.
Two objects of the same size may behave differently in water because their densities differ.
For example, a wooden log and an iron rod of the same size will have different densities; the iron rod is much denser and sinks while the wooden log floats.
Other factors such as shape and the way forces act (buoyancy) can also affect floating or sinking, but density is the primary property used to predict the behaviour.

What is density?



(a): Dense forest

(b): Less dense forest

Density is a measure of how much mass is contained in a unit volume of a substance.

Formal definition:

$$\text{Density} = \frac{\text{Mass}}{\text{Volume}}$$

Density does not depend on the shape or size of the sample but depends on the nature of the material and on temperature and pressure.

Temperature changes affect density because heating generally increases volume while mass remains the same; so density decreases on heating.

Pressure affects gases strongly (increasing pressure increases density by reducing volume); liquids and solids are much less compressible, so pressure has a negligible effect on their densities under ordinary conditions.

Oil packet example

Many 1-litre oil or ghee packets have a mass around 910 g. This indicates the density of the oil is less than 1 g/cm^3 (the density of water).

For example, 910 g in 1000 cm^3 gives a density of 0.91 g/cm^3, which explains why oil floats on water.

Units of density

The SI unit of density is **kilogram per cubic metre (kg/m³)**.

For laboratory and school work, densities of solids and liquids are often expressed in **gram per cubic centimetre (g/cm³)** or **gram per millilitre (g/mL)**; note that 1 mL = 1 cm³.

Useful conversions (written on a separate line):

$$1\ \mathrm{kg/m^3} = 0.001\ \mathrm{g/cm^3}$$

$$1\ \mathrm{g/cm^3} = 1000\ \mathrm{kg/m^3}$$

$$1\ \mathrm{mL} = 1\ \mathrm{cm^3}$$

At room temperature, 1 mL of water has a mass close to 1 g; therefore 10 mL ≈ 10 g, 100 mL ≈ 100 g, and so on, to a good approximation.

Relative density (specific gravity)

Relative density of a substance with respect to water is the ratio of its density to the density of water at the same temperature. It is a pure number with no units.

Example: If an aluminium block has mass 27 g and volume 10 cm³, its density is 2.7 g/cm³, so it is 2.7 times denser than water. Its relative density with respect to water is 2.7.

$$\text{Relative density of any substance with respect to water} = \frac{\text{Density of that substance}}{\text{Density of water at that temperature}}$$

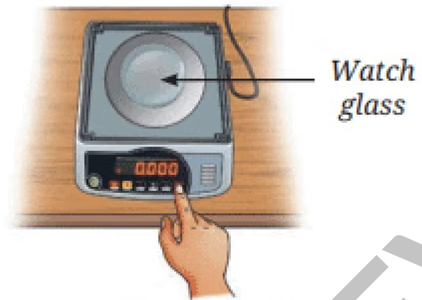
Determination of density: measuring mass and volume

Mass - how to measure

Mass is the quantity of matter in an object; the instrument used is a **balance**.

Digital balances give readings directly in grams or kilograms. Before measuring, ensure the balance shows zero; use the tare function to subtract the mass of a container.

(a): Digital weighing balance



(b): Tare the balance after placing a watch glass

(c): Weighing a solid object on digital balance

To find the mass of a solid, place a watch glass or small container on the pan, press tare (reset to zero), then place the solid and read the mass directly.
To measure the mass of a liquid, tare the empty beaker, then pour the liquid and read the mass shown.

Mass versus weight

Mass is the amount of matter, expressed in grams (g) or kilograms (kg).

Weight is the force of gravity acting on that mass and is measured in newtons (N). A balance normally measures weight but is calibrated to display mass units.

Volume - how to measure

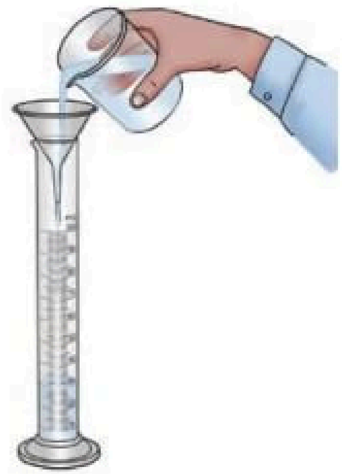
Volume is the space occupied by a substance. The SI unit is cubic metre (m^3), but in the laboratory we commonly use litre (L), millilitre (mL), or cubic centimetre (cm^3).

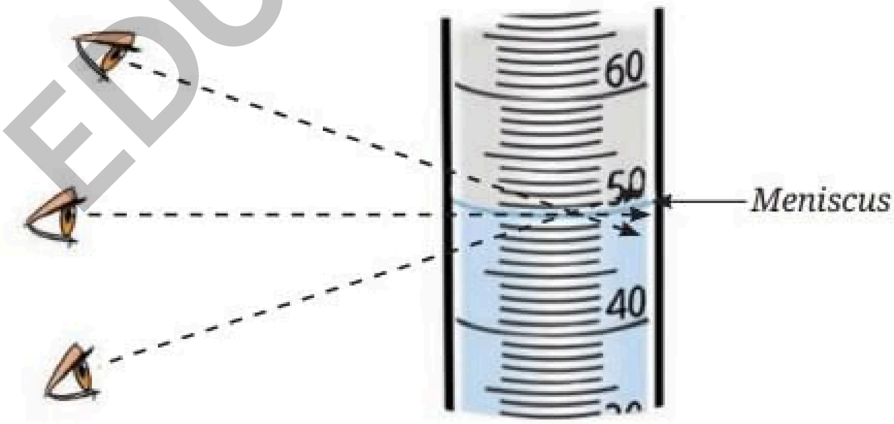
Volume of liquids is commonly measured using a **measuring cylinder** - a clear cylindrical tube marked with volume graduations.

Measuring cylinder of 100 mL

Measuring cylinders of different capacities

Choose the measuring cylinder closest in size to the required volume for best accuracy (for example use a 100 mL cylinder to measure 70 mL, rather than a much larger one).
Read the bottom of the meniscus (the curved surface of the liquid) at eye level for transparent liquids; for coloured liquids read the top of the meniscus.


Pouring water into the measuring cylinder



Measuring the reading

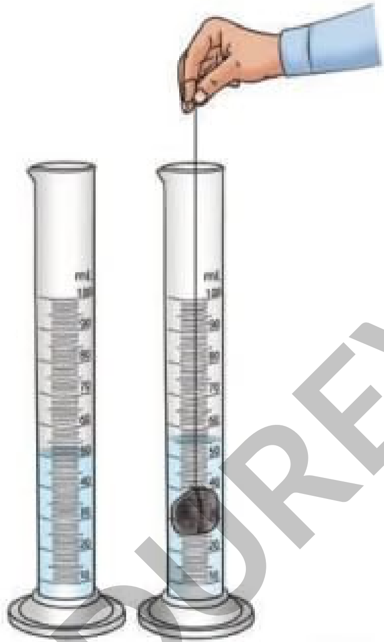
Determining volume of regular solids

For regular objects such as cuboids, use measurements with a scale or ruler and apply the formula **Volume = length × width × height**.

Example: A notebook with l = 25 cm, w = 18 cm and h = 2 cm has volume 25 × 18 × 2 = 900 cm^3.

Determining volume of irregular solids

Use the water displacement method in a measuring cylinder.



Level of water in the measuring cylinder (a) Without object; (b) With object

Fill the measuring cylinder with a known volume of water and note the reading.
Lower the irregular object (for example a stone tied to a thread) into the water and note the new level.
Volume of the object = Final water level - Initial water level. The units will be mL which equals cm^3.

Calculating density - worked example

Question: An object has a mass of 16.400 g and a volume of 5 cm^3. Calculate its density and express the answer in g/cm^3. Show working.

Solution:

Use the density formula.

$$\text{Density} = \frac{\text{Mass}}{\text{Volume}}$$

Substitute the given values on a new line.

$$\text{Density} = \frac{16.400 \text{ g}}{5 \text{ cm}^3}$$

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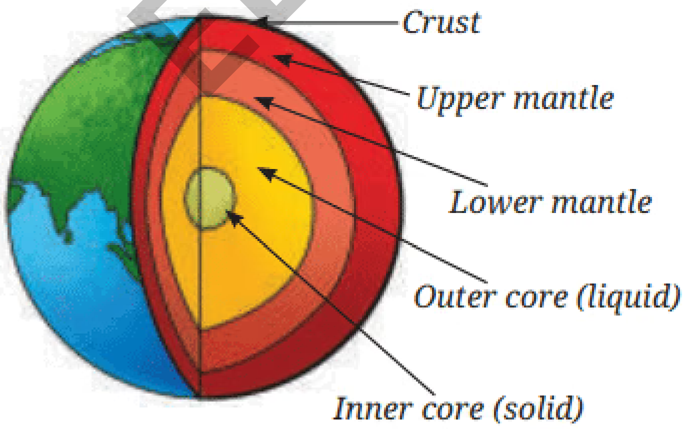
Carry out the division and express the final value on its own line.

$$\text{Density} = 3.28 \text{ g/cm}^3$$

Some interesting facts and examples

Earth is made of layers (crust, mantle, outer core, inner core) with differing densities; density generally increases with depth due to higher pressure and different composition.



Layers of Earth

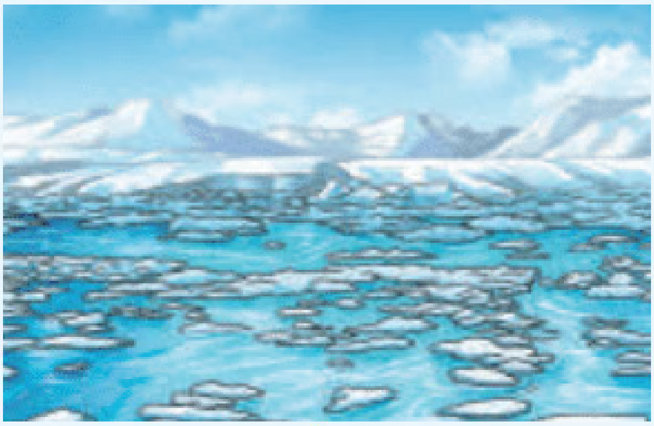
Bamboo and wooden logs were used in ancient times to make rafts and simple boats because they are light and float easily.

Bamboo raft floats on water

Hot air rises because heating decreases air density; hot air balloons operate using this principle.

Rising of hot air balloons

Why does ice float on water?



Ice floats on water

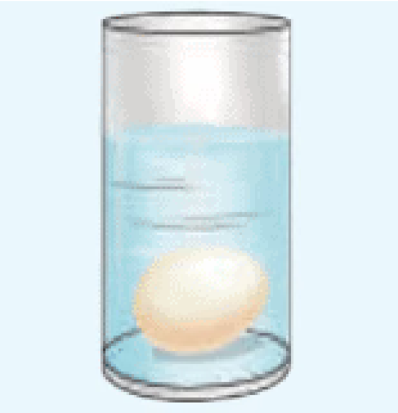
Water has the unusual property that its density is maximum at about 4 °C. On freezing (0 °C), water molecules form a crystalline structure that occupies more volume than liquid water. Because the same mass occupies a larger volume as ice, the density of ice is less than that of liquid water, and ice floats.

This behaviour is crucial to life in cold climates because a floating layer of ice insulates the water below, allowing fish and other organisms to survive.

Egg experiment - a practical observation

Place a fresh raw egg in a glass tumbler filled with tap water; the egg sinks to the bottom because its density is greater than that of fresh water.

By adding salt to the water and dissolving it, the density of the water increases and the egg may begin to float when the water's density becomes equal to or higher than the egg's density.


Raw whole egg sinks in water

Effect of temperature on density

In general, when a substance is heated its particles move further apart and the volume increases; because mass remains unchanged, density (mass/volume) decreases on heating.

This explains why warm air rises (it is less dense than surrounding cooler air) and why objects expand slightly on heating.

Effect of pressure on density

Pressure changes affect different states of matter differently.

Gases: Increasing pressure compresses gases, reducing their volume and increasing their density.

Liquids and solids: They are nearly incompressible under ordinary conditions, so pressure changes have only a very small effect on their densities.

Why ice floating matters and final notes

Because ice floats, it forms an insulating layer above liquid water during winter, which helps aquatic life survive in cold seasons.

Understanding solutes, solvents and solutions is useful across chemistry, biology, environmental science and everyday life - from cooking to medicine.

Key terms to remember

Solution: A uniform mixture formed when two or more substances mix evenly.

Solute: The substance that dissolves (in solid-liquid solutions, usually the solid).

Solvent: The substance that dissolves the solute (in solid-liquid solutions, usually the liquid).

Saturated solution: A solution that cannot dissolve more solute at a given temperature.

Unsaturated solution: A solution that can dissolve more solute at a given temperature.

Solubility: Maximum amount of solute that dissolves in a fixed amount of solvent at a specific temperature.

Concentration: Amount of solute present per unit volume of solution or solvent.

Density: Mass per unit volume, calculated using **Density = Mass / Volume**.

Relative density: Ratio of the density of a substance to the density of water (no units).

Device names: Balance (measures mass), measuring cylinder (measures liquid volume).

Temperature effects: Solubility of most solids in liquids increases with temperature; solubility of gases in liquids usually decreases with temperature.

You can practise questions from this chapter here: **Very Short Answer Questions: The Amazing World of Solutes, Solvents, and Solutions** (https://edurev.in/studytube/Very-Short-Answer-Questions-The-Amazing-World-of-Solutes--Solvents--and-Solutions/f5174c5c-9dad-46bf-8aea-ac457ddb7122_t)